

DATA ANALYTICS INTERNSHIP — ANALYTLAB AFRICA

One-Page Summary Report: Data Cleaning & EDA

Datasets: Online Retail | Netflix Titles | July 2026

DATASET DESCRIPTIONS

Dataset 1 — Online Retail: Transactional sales data from a UK-based online retailer covering December 2010 to December 2011. Contains 541,909 rows and 8 columns including InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

Dataset 2 — Netflix Titles: A catalogue of 8,807 movies and TV shows available on Netflix across 12 columns including title, type, director, cast, country, date_added, release_year, rating, duration, listed_in, and description.

CLEANING CHALLENGES ENCOUNTERED

1. Missing / Null Values

- Online Retail: 135,080 null CustomerID values (24% of dataset). Could not be imputed as CustomerID is an identifier column. Resolved by flagging with a CustomerID_missing column and filling with "Guest" placeholder.
- Netflix: Multiple columns contained nulls — director, cast, country, and rating. All were filled with "Unknown" to preserve row integrity.

2. Invalid Values

- Online Retail: Negative quantity values were found in the dataset representing returned or cancelled orders (invoices beginning with "C"). These distorted country-level bar charts by pulling bars below zero.
- Netflix: Release year validation found no values predating 1900 or exceeding 2023, however content from the 1940s was flagged as outliers in the box plot.

3. Inconsistent Records & Incorrect Formats

- Online Retail: InvoiceDate included full timestamps (e.g. 2010-12-01 08:26:00). A formatted_date column was created in YYYY-MM-DD format for consistency.
- Netflix: The date_added column was stored as a text string (e.g. "September 25, 2021") and required parsing using pd.to_datetime() with format "%B %d, %Y".
- Both datasets required column name standardisation — lowercased, stripped of spaces, and underscored.
- Text columns such as Description, Country, and Type were standardised to consistent casing using .str.upper() and .str.strip().

4. Outliers & Anomalies

- Online Retail: IQR analysis on Quantity flagged 58,501 rows as outliers — largely bulk wholesale orders of 24, 48, or 96 units. UnitPrice outliers flagged 39,450 rows driven by extreme high-value items and non-product entries such as POSTAGE and DISCOUNT.
- Netflix: No numeric outlier analysis was required beyond release_year. Box plot confirmed older titles (pre-1980) as lower-end outliers.

5. Summary Table Construction

- Building the pd.DataFrame() cleaning summary required careful tracking of all cleaning actions taken, records affected per column, and appropriate status labelling. The main challenge was ensuring Records Affected counts were drawn dynamically from the DataFrame rather than hardcoded.

CLEANING SUMMARY — BOTH DATASETS

Dataset	Issue Found	Action Taken	Status
Online Retail	Missing CustomerID (135,080 rows)	Flagged + filled with "Guest"	✓ Resolved
Online Retail	Duplicate rows	Removed with drop_duplicates()	✓ Resolved
Online Retail	Negative quantities (returns)	Separated into returns dataset	✓ Resolved
Online Retail	Non-product entries (POSTAGE, DISCOUNT)	Removed from main dataset	✓ Resolved

Online Retail	Outliers — Quantity & UnitPrice	Removed using IQR method	✓ Resolved
Online Retail	Inconsistent date format	Converted to YYYY-MM-DD	✓ Resolved
Online Retail	Text & column standardisation	Uppercased, stripped, renamed	✓ Resolved
Netflix	Missing director, cast, country, rating	Filled with "Unknown"	✓ Resolved
Netflix	date_added stored as text string	Parsed to YYYY-MM-DD format	✓ Resolved
Netflix	Inconsistent casing across text columns	Uppercased and stripped	✓ Resolved
Netflix	Column names inconsistent	Lowercased and underscored	✓ Resolved
Netflix	Duplicate rows	Removed with drop_duplicates()	✓ Resolved

KEY EDA FINDINGS

Online Retail Dataset

- Bar Chart: The United Kingdom dominates sales volume with over 3.75 million units — far exceeding all other countries combined.
- Histogram: Unit prices are heavily right-skewed with most products priced between £0 and £3, confirming a low-price, high-volume business model.
- Pie Chart: The UK contributes the largest share of total revenue across the top 10 countries.
- Line Chart: Seasonal quantity trends are visible with notable peaks during Q4, consistent with holiday shopping behaviour.
- Box Plot: Revenue distribution is concentrated at low values with significant high-value outliers representing bulk wholesale orders.

Netflix Titles Dataset

- Bar Chart: The United States produces significantly more Netflix content than any other country, followed by India and the United Kingdom.
- Histogram: Content is heavily skewed toward recent releases, with the majority of titles released between 2015 and 2020.
- Pie Chart: Movies outnumber TV Shows in the Netflix catalogue as a proportion of total content.
- Line Chart: Content additions peaked around 2019–2020 before dropping sharply in 2021, likely due to COVID-19 production disruptions.
- Box Plot: The median release year is approximately 2017, with older titles from the 1940s appearing as lower-end outliers.

TOP INSIGHTS

#	Dataset	Insight	Supporting Chart
1	Online Retail	The UK accounts for over 3.75M units sold — the business is overwhelmingly UK-dependent	Bar Chart
2	Online Retail	24% of all transactions are from guest/unidentified customers, limiting customer-level analysis	Data Cleaning

3	Online Retail	Most products are priced under £3 — this is a low-cost, high-volume retail operation	Histogram
4	Online Retail	Negative quantity values from returns distort geographic analysis if not separated first	Bar Chart
5	Online Retail	Bulk wholesale orders (24–96 units) represent a significant customer segment	Box Plot
6	Netflix	The USA dominates Netflix content production — the platform remains US-centric despite global expansion	Bar Chart
7	Netflix	Movies outnumber TV Shows but TV Shows drive longer platform engagement per title	Pie Chart
8	Netflix	Netflix content additions peaked in 2019–2020 and dropped sharply in 2021 due to COVID-19	Line Chart
9	Netflix	Most Netflix content was released after 2015 — the platform favours recent productions	Histogram
10	Netflix	Older classic titles (pre-1980) exist as outliers, retained to serve niche audiences	Box Plot